Influence of a Cluster Set Configuration on the Adaptations to Short-Term Power Training

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Abstract

Morales-Artacho, AJ, Padial, P, García-Ramos, A, Pérez-Castilla, A, and Feriche, B. Influence of a cluster set configuration on the adaptations to short-term power training. *J Strength Cond Res* 32 (4): 930–937, 2018—This study investigated the effects of a traditional (TT) vs. cluster (CT) resistance training on the lower-body force, velocity, and power output. Nineteen males were allocated to a CT or a TT group and took part of a 3-week resistance training (2 weekly sessions). CT involved 6 sets of 3 × 2 repetitions (30 seconds rest every 2 repetitions and 4 minutes 30 seconds between sets). TT comprised 6 sets of 6 continuous repetitions (5 minutes rest between sets). Before and after the training period, force (F_{25}, F_{50}, F_{75}), velocity (V_{25}, V_{50}, V_{75}), and power (P_{25}, P_{50}, P_{75}) were obtained during the countermovement jump (CMJ) exercise at 3 external loading conditions (25, 50, and 75% of body mass). Individual linear regressions were used to determine the force–velocity profile including the *Slope*, estimated maximal theoretical force (F_0), velocity (V_0), and power (P_0). After CT, *very-likely moderate* increments in P_{25} were observed compared with TT ($p = 0.011$, ES = 0.55) because of a *very-likely moderate* rise in V_{25} ($p = 0.001$, ES = 0.71). No significant differences were observed in any of the *F-v* profile variables between the TT and CT groups ($p \geq 0.207$, ES $\leq$ 0.31). Our results suggest that 3 weeks of muscle power training including cluster set configurations are more efficient at inducing velocity and power adaptations specific to the training load.

Key Words resistance training, force–velocity relationship, muscle strength, muscle power

Introduction

The capacity to generate power output is a key performance factor in sport disciplines that involve explosive actions such as sprinting, throwing, or kicking (22). Indeed, evidence from cross-sectional studies shows that the ability to produce high levels of power may characterize athletes' level and sport modalities (22). Accordingly, the optimization of training interventions aiming to improve maximal power output has been for a long time a relevant topic of research in the strength training field.

Under isoinertial conditions, force, velocity, and mechanical power outputs decrease when continuous repetitions are performed at maximal intended velocity (21). For example, during bench press and squat exercises, unintentional decrements in movement velocity have been shown to occur as neuromuscular fatigue develops (30). It is thus commonly proposed that fatigue should be minimized and training intensity maximized (i.e., movement velocity) when training for maximal neuromuscular adaptations (13,27). In this context, evidence-based training guidelines suggest limiting the number of repetitions performed per set (i.e., 2–5 repetitions) (1) and allowing enough between-set rest periods (i.e., from 3 to 5 minutes) (8). Additionally, appropriate manipulation of the training sets configuration (i.e., implementing resting periods between repetitions) seems to be useful at enhancing mechanical variables and potentially improving power training adaptations (12).

The introduction of short rest intervals (15–30 seconds) between a group or cluster of repetitions within a set (i.e., cluster training) acutely improves power performance during commonly used strength training exercises (12,18). Compared with traditional set configurations, the squat-jump (14), ballistic bench-press (9), and clean exercises (16) have been shown to benefit from cluster loading configurations (i.e., higher power output and movement velocity). On this basis, it is suggested that ballistic explosive training comprising cluster sets may lead to greater performance adaptations than training periods involving traditional set configurations (12). However, despite the known acute benefits, very few studies have investigated the chronic effects of cluster loadings on power training adaptations. While training with continuous repetitions seems to favor maximal strength gains (15,24), there is conflicting and limited evidence on the effects of cluster training on maximal power development. Lawton et al. (24) did not find superior power output development using a cluster set configuration after 6 weeks of bench press training. On the contrary, after an 8-week resistance training intervention on elite rugby players, Hansen et al. (15) described positive effects of cluster loading on peak

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power and movement velocity during the squat jump exercise. Consequently, the effects of cluster set loading on training power adaptations are unclear and further research is clearly needed. Furthermore, the study of the force–velocity relationship may help to gain further insight into the effects of cluster training on the muscle mechanical capabilities (19,29).

The purpose of the present study was to investigate the effects of 2 different muscle power training set configurations (traditional vs. cluster) on the lower-body maximal force, velocity, and power capabilities. It was hypothesized that a power training period comprising cluster sets would lead to further short-term power performance improvements.

Methods

Experimental Approach to the Problem

A randomized repeated measures design, including intragroup and intergroup comparisons, was employed to evaluate the effects of set configuration during a short-term power-oriented resistance training mesocycle on maximal

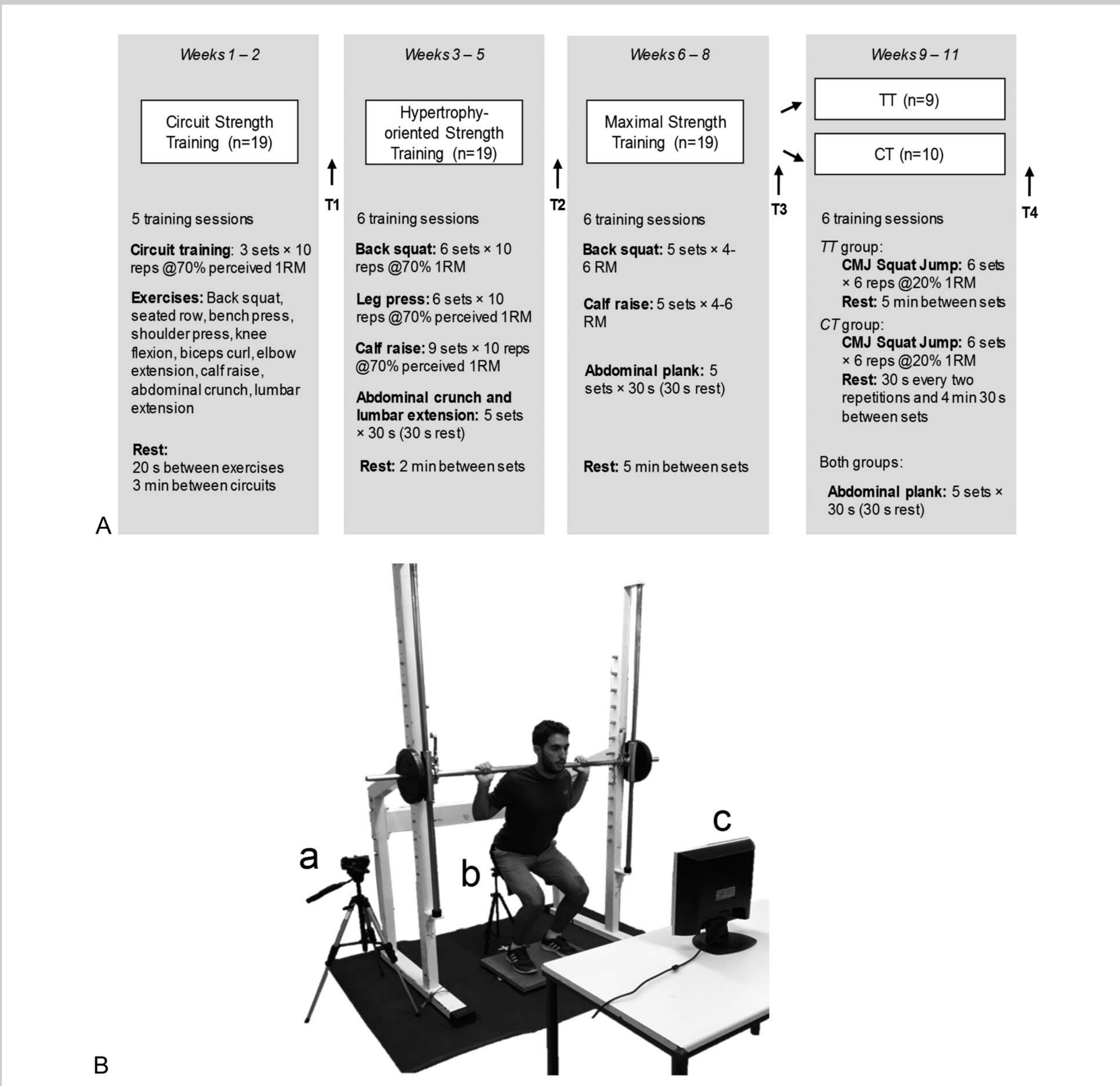

Figure 1. Training program and experimental setup. A) Exercises, sets, repetitions (reps), and intensities (% 1RM) prescribed in the entire resistance training intervention. Training sessions were performed twice weekly. B) Experimental setup including video camera (a), screen (c), and adjustable rod on a tripod (b) used to control squat depth.

force, velocity, and power output. Before the power training intervention, subjects took part in a general preparatory phase aiming to increase strength levels (6,32).

Specifically, subjects performed 3 mesocycles (8 weeks) of progressive resistance training including 2 weeks of strength circuit training, 3 weeks of hypertrophy-oriented strength training, and 3 weeks of maximal strength training. Training exercises, volumes, and intensities are shown in Figure 1A. The individual force-velocity (F-v) relationship, determined by an incremental loading test at 3 different external loads relative to their body weight (BW; 25, 50 and 75% BW), and the maximal strength levels (1RM) of the lower limbs, were assessed before and after the corresponding cluster (CT) or traditional (TT) resistance training period. The same variables were also evaluated at the end of each mesocycle of the general phase for further comparisons. All evaluations were conducted at the same time of the day (±1 hour) for each subject and under similar environmental conditions (~22° C and ~60% humidity).

Subjects

Nineteen male volunteered to participate in this study. All subjects were physically active Sport Science students familiarized with the resistance training exercises employed, with no previous systematic strength training background. Body mass, height, and age at the start of the study were ± *SD* 76.6 ± 9.7 kg, 178.3 ± 6.3 cm and 23.6 ± 5.8 years, respectively. Subjects did not report any physical limitations, health problems, or musculoskeletal injures that could compromise testing. They were also instructed to avoid any strenuous exercise over the course of the study. All volunteers were informed regarding the nature, aims, and risks associated with the experimental procedures before they gave their written consent to participate. This study was approved by the University of Granada and conformed to the standards of the Declaration of Helsinki.

Procedures

Testing Protocol. Subjects were instructed to maintain their habitual habits throughout the study. All testing sessions were performed between 11 and 20 hours, at least 72 hours after the last training session. Each subject was always tested at a consisted time of day (±1 hour). Subject's weight (BF 350; Tanita Tokyo, Japan) and height (Seca Instruments Ltd., Hamburg, Germany) were measured at the start of each testing session. Subjects performed a standardized warm-up protocol consisting of 5-minute cycling, joint mobility exercises, and 3–4 unloaded countermovement jumps (CMJs). Afterward, they completed an incremental loading CMJ test (load accuracy ≥0.25 kg) at 25, 50, and 75% of their own BW in a Smith Machine (Technogym, Gambettola, Italy). Instructions were given to keep constant downward pressure on the barbell and to jump as high as possible. At each load, 2 attempts were performed and resting time between both repetitions was set to 1 minute. The repetition with the highest takeoff velocity was selected for further analysis. To minimize variation on jump kinematic and kinetic patterns (26), a manual goniometer and an adjustable rod on a tripod were used to set CMJ depth to 90° knee angle for each subject. Subjects were instructed to squat until touching the rod with their glutei. Additionally, visual biofeedback was provided during each jump using a video camera placed laterally to subjects' position (Figure 1B).

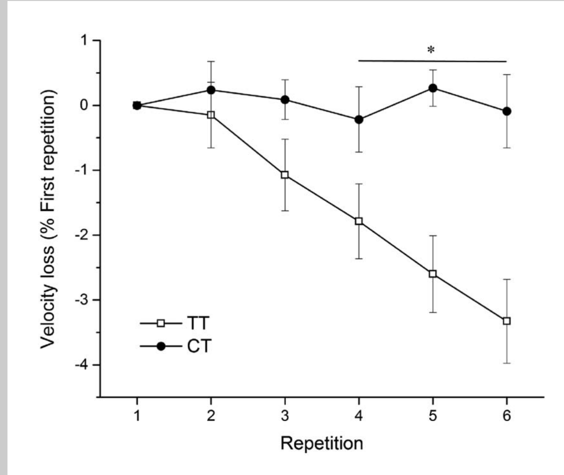

Figure 2. Velocity loss during the power training sessions. Peak velocity percentage changes from the first repetition. The figure represents all repetitions averaged across the 6 power training sessions from the traditional (TT) and cluster training (CT) groups. Data are mean ± *SEM*. *$p < 0.001$, ES ≥ 0.94.

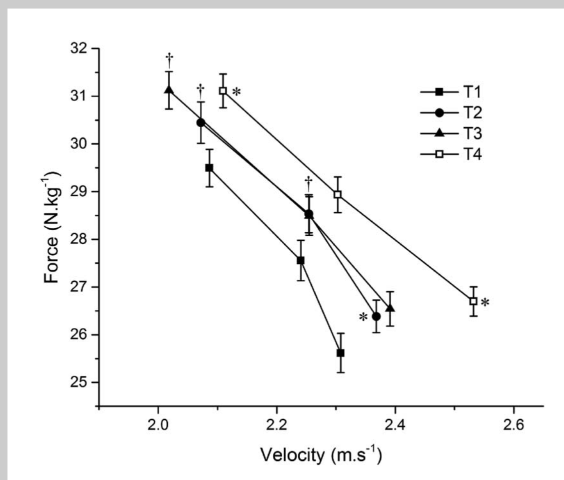

Figure 3. Force–velocity curves after the introductory mesocycle (2 weeks of circuit weight training [T1]), general strength mesocycle (3 weeks of hypertrophy-oriented resistance training [T2]), maximal strength mesocycle (3 weeks of maximal resistance training [T3]), and especial strength mesocycle (3 weeks of power-oriented resistance training [T4]) (all subjects, *n* = 19). *Significant increase in peak velocity compared with the previous test ($p < 0.05$). †Significant increase in peak force from previous test ($p < 0.05$).

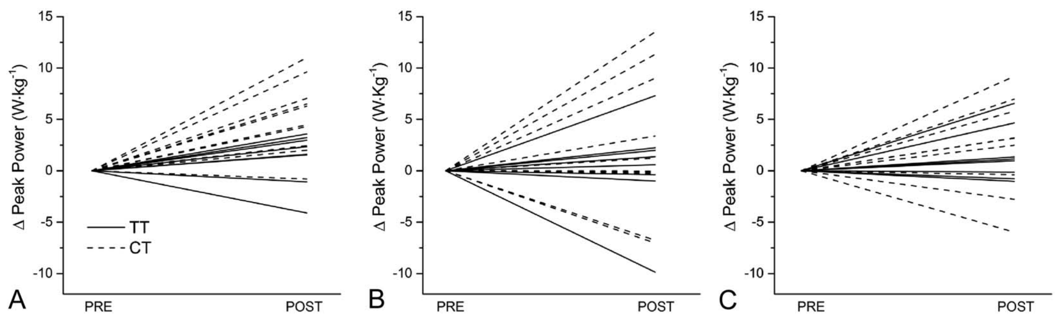

Figure 4. Individual absolute changes in peak power (W·kg^{-1}) at 25% (A), 50% (B) and 75% (C) body weight (BW) loading conditions after 3 weeks of traditional (TT, solid lines) and cluster (CT, dashed lines) resistance training.

Afterward, subjects completed a maximal dynamic strength test (1RM) during the half squat exercise (90° knee angle) on the same Smith Machine. Three sets of 3–6 repetitions at increasing loads were completed before performing one set of 2–3 repetitions to failure. Rest periods between sets were kept to 5 minutes. Squat depth was also standardized as described above and visual feedback from the video camera laterally positioned was also provided. Spotters verbally encouraged subjects throughout all lifts. Using the 2–3RM load, 1RM values were predicted using Brzycki's equation (4).

Data Collection and Processing. All jumps were performed on a force plate (Type 9260AA6; Kistler, Winterthur, Switzerland). Vertical ground reaction force (GRF) data were recorded at 1,000 Hz using an analog-to-digital converter (DAQ system 5691A1; Kistler) and collected through the BioWare software (Kistler). Data analysis was performed using custom written scripts computed with Matlab software (Version R2015a; The Mathworks, Natick, MA, USA). Baselines were carefully checked and a 1-second averaged GRF period was selected as baseline on each jump. The onset of each CMJ was taken when GRF was 10 N below the baseline. Force, velocity, and power data were calculated from the GRF F_z component, according to the impulse-momentum approach (25). Specifically, vertical instantaneous acceleration (a) of the center of mass was first calculated using the following equation:

$$a = \frac{F}{m} - g$$

Where m is the total body mass (BM; kg).

Acceleration was then integrated once to provide instantaneous vertical velocity (v):

$$v = \int a \, \mathrm{d}t + V_0$$

Given that each CMJ attempt started from static conditions, at t_0:

$$v_0 = 0$$

Power (in Watts) was then calculated as the product of force and velocity ($P = F \cdot v$). Peak force (F_{25}, F_{50}, F_{75}), velocity (V_{25}, V_{50}, V_{75}), and power (P_{25}, P_{50}, P_{75}) data were further analyzed at each loading condition.

Force–Velocity Profiling. For each participant, the force–velocity relationship obtained from the peak force and

Table 1. Effects of the 3-week hypertrophy training period on peak power.*

	T1 (W·kg^{-1})	T2 (W·kg^{-1})	%Δ	ES (90% CI)	% chances†
P_{25}	49.89 ± 5.97	53.45 ± 6.28‡	+7.31 ± 6.56	0.57 (0.37 to 0.77)	100/0/0
P_{50}	53.42 ± 7.38	55.94 ± 7.42§	+5.10 ± 8.71	0.33 (0.09 to 0.56)	82/18/0
P_{75}	54.12 ± 9.18	55.67 ± 7.84	+3.87 ± 11.51	0.16 (−0.09 to 0.41)	39/60/1

*Peak power at 25% (P_{25}), 50% (P_{50}) and 75% (P_{75}) of body weight loads during Test 1 (T1) and Test 2 (T2).
†% chances of better/trivial/worse effect.
‡$p = 0.001$.
§$p = 0.028$.

Table 2. Effects of the 3-week maximal strength training period on peak power.*†

	T2 ($W \cdot kg^{-1}$)	T3 ($W \cdot kg^{-1}$)	%Δ	ES (90% CI)	% chances
P_{25}	53.45 ± 6.28	54.30 ± 6.71	+1.63 ± 4.79	0.14 (−0.03 to 0.30)	25/75/0
P_{50}	55.94 ± 7.42	56.02 ± 8.45	+0.31 ± 9.92	0.01 (−0.27 to 0.29)	12/78/10
P_{75}	55.67 ± 7.84	55.03 ± 9.54	−1.08 ± 10.76	−0.07 (−0.31 to 0.18)	4/78/18

*Peak power at 25% (P_{25}), 50% (P_{50}) and 75% (P_{75}) of body weight loads during test 3 (T3) and test 4 (T4).
†% chances of better/trivial/worse effect.

velocity values was modeled by a least-squares linear regression. The intercepts of the *F-v* curve with the force and velocity axis were taken to obtain the theoretical maximal force (F_0) and maximal velocity (V_0). The slope of the *F-v* relationship and maximal theoretical power output (P_0, computed as $P_0 = [F_0 \cdot V_0]/4$) (28) were also obtained for each participant.

Training Program. Throughout the course of the intervention, subjects refrained from any additional lower-body strength training activity. Subjects' attendance to training sessions was 100% and all sessions were supervised. The 8-week resistance training period before power training ensured that all subjects improved their strength levels and exercises technique. Training volume, intensity, and resting periods during each corresponding training period are shown in Figure 1. Training sessions started with a standardized warm-up protocol consisting of 5 minutes cycling, 3–5 minutes of lower-body mobility exercises, 10 back squat repetitions with an unloaded 17 kg barbell, and 6 back squat repetitions with a load corresponding to 50% of the correspondent session training intensity. Additionally, during the 3-week power training period, 5 unloaded CMJs and 5 loaded CMJs with a 17-kg barbell were added to the warm-up protocol. Training intensity was adjusted on each training phase relative to the latest 1RM assessment performed. All training sessions were completed twice weekly with at least 48 hours rest between them. During the power training phase, subjects on both TT and CT groups were provided with velocity feedback from a linear transducer during all training sessions (1,000 Hz; Globus Italia, Codognè, Italy). Specifically, barbell peak velocity from each repetition was screened and verbal encouragement was given to achieve maximal peak velocity.

Statistical Analyses

Both clinical magnitude-based inferences (2) and the null-hypothesis significance testing were performed to assess changes in the dependent variables. Data are presented as mean ± *SD*. For presentation purposes, data within figures are mean ± *SEM*. The normality of the data was tested using a Shapiro–Wilk's test. Between-group differences at

Table 3. Pre-to-post changes in power, velocity, and force at each loading condition after the traditional and cluster training periods.

			Between-group differences		
	TT group (T4 − T3 Δ change)	CT group (T4 − T3 Δ change)	ES (90% CI)	% chances*	Interaction effect (p, ANOVA)
P_{25} ($W \cdot kg^{-1}$)	+1.21 ± 2.45	+5.29 ± 3.57	0.55 (0.22 to 0.88)	96/4/0	0.011
P_{50} ($W \cdot kg^{-1}$)	+0.27 ± 4.50	+2.42 ± 7.01	0.23 (−0.27 to 0.74)	54/38/8	0.444
P_{75} ($W \cdot kg^{-1}$)	+1.43 ± 2.55	+3.29 ± 5.91	0.18 (−0.17 to 0.53)	46/51/4	0.395
V_{25} ($m \cdot s^{-1}$)	+0.06 ± 0.09	+0.22 ± 0.10	0.71 (0.37 to 1.05)	99/1/0	0.001
V_{50} ($m \cdot s^{-1}$)	+0.01 ± 0.17	+0.08 ± 0.24	0.24 (−0.35 to 0.84)	55/34/11	0.494
V_{75} ($m \cdot s^{-1}$)	+0.06 ± 0.09	+0.12 ± 0.17	0.22 (−0.14 to 0.58)	53/44/3	0.305
F_{25} ($N \cdot kg^{-1}$)	+0.43 ± 0.96	−0.09 ± 1.73	−0.30 (−0.95 to 0.34)	9/30/61	0.616
F_{50} ($N \cdot kg^{-1}$)	+0.71 ± 0.84	0.21 ± 1.27	−0.26 (−0.70 to 0.19)	5/36/59	0.333
F_{75} ($N \cdot kg^{-1}$)	+0.18 ± 0.62	−0.18 ± 1.09	−0.19 (−0.55 to 0.18)	4/47/49	0.394

*% chances of better/trivial/worse effect.

Table 4. Pre-to-post changes in the *F-v*-derived parameters after the traditional and cluster training periods.*

	Traditional training		Cluster training		Between-group differences		
	T3	T4	T3	T4	ES (90% CI)	% chances†	Interaction effect (p, ANOVA)
F_0 ($N \cdot kg^{-1}$)	47.97 ± 10.44	50.59 ± 19.19	51.87 ± 14.3	50.33 ± 9.22	−0.31 (−1.17 to 0.56)	16/26/58	0.537
V_0 ($m \cdot s^{-1}$)	5.51 ± 1.23	5.94 ± 1.60	5.73 ± 1.30	6.08 ± 1.34	−0.06 (−0.73 to 0.61)	25/40/36	0.870
P_0 ($W \cdot kg^{-1}$)	63.95 ± 8.89	69.61 ± 8.61	71.56 ± 11.19	73.99 ± 8.28	−0.27 (−0.65 to 0.10)	2/34/63	0.207
Slope	−9.41 ± 3.99	−10.09 ± 7.88	−9.91 ± 5.24	−9.09 ± 4.34	0.29 (−0.66 to 1.25)	57/24/19	0.593

*Maximal theoretical force (F_0), velocity (V_0), and power (P_0) and slope of the *F-v* relationship before and after the training intervention. No significant interaction effects were revealed by the ANOVA.
†% chances of better/trivial/worse effect.

baseline were tested using independent samples Student's t tests. Paired Student's t tests were employed to assess changes on maximal strength and peak power after the hypertrophy and maximal strength training periods. A 2-way repeated measures ANOVA was used to evaluate the influence of group (CT vs. TT) and repetition number (1 vs. 2 vs. 3 vs. 4 vs. 5 vs. 6) on the peak velocity percentage change from first repetition averaged across all power training sessions. Post hoc tests were performed by means of Bonferroni procedures when appropriate. A 2-way repeated measures ANOVA was employed to assess the influence of time (pre vs. post) and group (TT vs. CT) on each dependent variable. The level of significance was set at $p < 0.05$.

Cohen's d effect sizes (ES) (90% CI) and thresholds (< 0.5 [small]; 0.5–0.79 [moderate]; ≥0.8 [large]) were also used to compare the magnitude of the differences. The smallest worthwhile change in the mean in standardized (Cohen) units was set to 0.2 of the between-subject *SD*. Quantitative chances of better or worse effects (i.e., greater than the smallest worthwhile change) were assessed qualitatively as follows: <1%, almost certainly not; 1–5%, very unlikely; 5–25%, unlikely; 25–75%, possible; 75–95%, likely; 95–99%, very likely; and >99%, almost certain. If the chances of having beneficial or detrimental were both >10%, the true difference was assessed as unclear. Magnitude-based inferential statistics were calculated using a modified built-in spreadsheet (17).

Results

At the start of the training intervention, there were no differences between groups in BW ($p = 0.574$) and 1RM values ($p = 0.813$). No changes in BW were observed during the training intervention (75.2 ± 9.6 vs. 75.6 ± 9.9 kg; $p > 0.05$).

Large improvements in maximal strength levels (1RM) were observed from T1 to T2 (2.11 ± 0.31 vs. 2.47 ± 0.30 $kg \cdot kg^{-1}$; $p < 0.001$, ES = 1.18) and from T2 to T3 (2.47 ± 0.30 vs. 2.70 ± 0.33 $kg \cdot kg^{-1}$; $p < 0.001$, ES = 0.72) in all subjects. ANOVA showed no significant main effect of time (2.70 ± 0.33 vs. 2.74 ± 0.35 $kg \cdot kg^{-1}$; $p = 0.16$, ES = 0.1) or time × group interaction from T3 to T4 (2.65 ± 0.37 vs. 2.72 ± 0.41 $kg \cdot kg^{-1}$ for TT, and 2.75 ± 0.30 vs. 2.76 ± 0.31 $kg \cdot kg^{-1}$ for CT; $p = 0.32$, ES = 0.05).

Changes in P_{25}, P_{50} and P_{75} after the 3-wk moderate and the 3-wk maximal strength training periods are shown in Tables 1 and 2, respectively.

The ANOVA showed a *large* (ES = 1.01) significant condition × repetition interaction effect ($p < 0.001$) because of greater decreases in peak velocity at repetitions 4, 5, and 6 during TT ($p \leq 0.02$; ES ≥ 0.94) (Figure 2).

The *F-v* relationship changes throughout the training intervention including all subjects are presented in Figure 3.

A significant main effect for time revealed that 3 weeks of power training incremented P_{25} (54.29 ± 6.71 vs. 57.65 ± 7.07 $W \cdot kg^{-1}$, ANOVA, $p < 0.001$, ES = 0.49) and P_{75}

(55.03 ± 9.54 vs. 57.44 ± 8.74 W·kg^{-1}, ANOVA, $p = 0.041$, ES = 0.26) because of *moderate* increments in V_{25} (2.39 ± 0.21 vs. 2.53 ± 0.21 m·s^{-1}, ANOVA, $p < 0.001$, ES = 0.67) and *small* increases V_{75} (2.02 ± 0.27 to 2.11 ± 0.25 m·s^{-1}, ANOVA, $p = 0.012$, ES = 0.35). Time × group interaction effects (ANOVA) in force, velocity, and power are shown in Table 3. Individual changes in peak power at each loading condition after both traditional and cluster training are shown in Figure 4.

ANOVA denoted no significant main effects of time for F_0 (49.92 ± 12.37 vs. 50.45 ± 15.05 N·kg^{-1}; $p = 0.34$; ES = 0.05), V_0 (5.61 ± 1.26 vs. 6.00 ± 1.47 m·s^{-1}; $p = 0.14$, ES = 0.41) or *Slope* (-9.66 ± 4.64 vs. -9.59 ± 6.36 N·s·m^{-1}; $p = 0.95$, ES = -0.01). There was a significant main effect of time for P_0 (67.75 ± 10.11 vs. 71.80 ± 2.11 W·kg^{-1}; $p = 0.005$, ES = 0.55). No significant time × group interaction effects were found in any of the *F-v* profile variables (Table 4).

Discussion

The current study aimed to evaluate the influence of 2 different set configurations on the short-term muscle power training adaptations. Accordingly, the effects of a 3-week power training period comprising TT or CT on strength and power output were investigated. Compared with TT, greater improvements in peak power and velocity output were observed after CT. These results seem to support previous suggestions on the efficacy of CT at inducing short-term velocity and power adaptations after ballistic training.

In agreement with our hypothesis, the CT group showed greater improvements in power output (9.7 vs. 2.7% in P_{25}) and movement velocity (8.2 vs. 2.3% in V_{25}) than the TT group although no clear differences were observed in the *F-v* profile. These results support previous suggestions on the effectiveness of cluster set configurations for explosive ballistic training (12) and the lack of differences on the *F-v* profile (19). Hansen et al. (15) also described greater velocity and power adaptations after CT in elite athletes although these improvements were not consistent across the loading conditions tested. Similarly, we observed greater power and velocity improvements after CT at low loading conditions, but no meaningful differences were detected at higher loading conditions. Load-specific training adaptations may therefore have been responsible for these results (5). The power training load employed in the present study was velocity-oriented (20% 1RM) and performance was consequently maximized mostly at light loading conditions. Additional research should examine whether cluster set configurations are also beneficial at eliciting greater adaptations when training at different loading conditions (i.e., maximal strength or hypertrophy-oriented training).

Consistent with previous studies (9,31), the barbell velocity data registered during the power training sessions showed that the inclusion of a 30-second rest every 2 repetitions effectively maximized training intensity (Figure 2), which may have influenced the mechanisms of adaptation during CT. Interestingly, earlier investigations have shown that cluster set configurations lead to faster phosphocreatine replenishment rates (11), lower acid lactic accumulation (10), and lower ratings of perceived exertion (16,20). Given that the actual movement velocity (27) and the intention to move explosively (3,19) are known to be critical factors when targeting neuromuscular adaptations, CT may have provided optimal conditions for power training adaptations.

Regarding maximal strength gains, no significant changes were observed after the TT or CT periods (+2.5 vs. +1.3%, respectively). Training interventions involving continuous set configurations have been proposed to result in greater upper-body (24,28) and lower-body maximal strength gains and force application while jumping (15), although this is not always the case (20). Given the velocity-oriented training load employed in our intervention, and considering the improvements in 1RM (+21.7%) after the previous 6-week sequential resistance training period, no substantial changes in maximal strength levels were expected after the power training period. Notwithstanding, despite no meaningful maximal strength improvements after the power training phase, greater force application during CMJ at 50% BM and a tendency toward "small" improvements in F_0 (+3.2%; ES = 0.31) and P_0 (+9.2%; ES = 0.27) after TT compared with CT were observed. These findings further support previous research describing greater increments in force application during CMJ after lower-body traditional power training (15).

Our results suggest that a cluster set configuration may allow to achieve superior velocity and power performance adaptations at the specific training loading condition after short-term CMJ training. Nevertheless, several facts must be considered when interpreting these findings. First, it should be taken into account that although subjects were randomly assigned to CT or TT group based on their maximal strength levels (6), the *F-v* profile imbalance may also be appropriate to be used for training group allocation, which could yield different training effects of cluster loadings on the *F-v* profile (23). Furthermore, it should be kept in mind that the study sample consisted of active non–resistance-trained subjects. Adaptations may be influenced by training background and the current results should be applied with caution to strength-trained populations.

Practical Applications

The inclusion of a 30-second resting period every 2 CMJ repetitions performed at the maximal intended velocity allows not only for acute improvements on movement velocity, but it also affects the short-term power training responses. Strength and conditioning coaches may thus consider using cluster loading configurations to facilitate short-term power training performance adaptations. In this context, cluster loading may be useful to be employed during tapering periods aiming to maximize power muscle capabilities (7). These findings support the idea that maximizing movement velocity and training intensity by including intraset rest periods positively influences the short-term power training adaptations.

Acknowledgments

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